

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet.

range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. Subnetting 192.168.20.0/24 for Three Labs

Given:

- Network: 192.168.20.0/24
- Lab A: /26
- Lab B: /27
- Lab C: /28

Step 1: Calculate subnet details

Lab	Prefi x	Subnet Mask	Total Addresses	Usable Hosts
Lab A	/26	255.255.255.1 92	64	62
Lab B	/27	255.255.255. 24	32	30
Lab C	/28	255.255.255.2 40	16	14

Lab A – /26

Block size:

$$256 - 192 = 64$$

Network address = **192.168.20.0**

- Network Address: 192.168.20.0
- Valid Host Range: 192.168.20.1 – 192.168.20.62
- Broadcast Address: 192.168.20.63
- Usable Hosts: 62

Lab B – /27

The next available block starts at 192.168.20.64.

Block size:

$$256 - 224 = 32$$

- Network Address: 192.168.20.64
- Valid Host Range: 192.168.20.65 – 192.168.20.94
- Broadcast Address: 192.168.20.95

- Usable Hosts: **30**

Lab C – /28

The next available block starts at **192.168.20.96**.

Block size:

$$256 - 240 = 16$$

- Network Address: **192.168.20.96**
- Valid Host Range: **192.168.20.97 – 192.168.20.110**
- Broadcast Address: **192.168.20.111**
- Usable Hosts: **14**

Final Answer:

Lab	Network Address	Valid Host Range	Broadcast	Usable Hosts
A	192.168.20.0/26	192.168.20.1 – 62	192.168.20.63	62
B	192.168.20.64/27	192.168.20.65 – 94	192.168.20.95	30
C	192.168.20.96/28	192.168.20.97 – 110	192.168.20.111	14

Conclusion: The three labs are assigned separate subnets without overlapping IP addresses. The total usable host capacity is **106 hosts**.

2. Subnet Analysis of Two Systems

Given:

- System 1 IP: **192.168.1.130**
- System 2 IP: **192.168.1.190**
- Subnet Mask: **255.255.255.192**
- Prefix: **/26**

Step 1: Determine block size

The last octet of the subnet mask is **192**.

$$256 - 192 = 64$$

Therefore, the **/26** subnets are:

- 192.168.1.0 – 63
- 192.168.1.64 – 127
- 192.168.1.128 – 191
- 192.168.1.192 – 255

System 1: 192.168.1.130

The IP address 130 falls between 128 and 191.

Therefore:

- Subnet Range: 192.168.1.128 – 192.168.1.191
- Network Address: 192.168.1.128
- Valid Host Range: 192.168.1.129 – 192.168.1.190
- Broadcast Address: 192.168.1.191

The given IP 192.168.1.130 is within the valid host range.

Result: System 1 IP is valid for its subnet.

System 2: 192.168.1.190

The IP address 190 also falls between 128 and 191.

Therefore:

- Subnet Range: 192.168.1.128 – 192.168.1.191
- Network Address: 192.168.1.128
- Valid Host Range: 192.168.1.129 – 192.168.1.190
- Broadcast Address: 192.168.1.191

The given IP 192.168.1.190 is within the valid host range.

Result: System 2 IP is valid for its subnet.

Usable Host Calculation:

For /26:

$$2^{32-26}-2=2^6-2=64-2=62$$

Final Answer:

System	IP Address	Network	Valid Host Range	Broadcast	Valid ?
System 1	192.168.1.130	192.168.1.128	.129 – .190	.191	Yes
System 2	192.168.1.190	192.168.1.128	.129 – .190	.191	Yes

Conclusion: Both systems belong to the **same /26 subnet** and can communicate directly at Layer 2 without requiring routing, assuming they are connected to the same network segment. The subnet provides **62 usable host addresses**.

3. Subnet Allocation for Organization Departments

Given:

Network Block = 172.16.0.0/16

Departments:

- HR → /25
- Sales → /26
- IT → /27
- Admin → /28

Since different subnet sizes are required, the allocation is performed sequentially using the specified subnet masks.

HR Department – /25

Subnet Mask:

255.255.255.128

Block size:

256-128=128

Allocation:

- Network Address: 172.16.0.0
 - Valid Host Range: 172.16.0.1 – 172.16.0.126
 - Broadcast Address: 172.16.0.127
 - Usable Hosts: 126
-

Sales Department – /26

Subnet Mask:

255.255.255.192

Block size:

$256 - 192 = 64$

Next available network = 172.16.0.128

Allocation:

- Network Address: 172.16.0.128
 - Valid Host Range: 172.16.0.129 – 172.16.0.190
 - Broadcast Address: 172.16.0.191
 - Usable Hosts: 62
-

IT Department – /27

Subnet Mask:

255.255.255.224

Block size:

$256 - 224 = 32$

Next available network = 172.16.0.192

Allocation:

- Network Address: 172.16.0.192
 - Valid Host Range: 172.16.0.193 – 172.16.0.222
 - Broadcast Address: 172.16.0.223
 - Usable Hosts: 30
-

Admin Department – /28

Subnet Mask:

255.255.255.240

Block size:

256-240=16 256-240=16 256-240=16

Next available network = 172.16.0.224

Allocation:

- Network Address: 172.16.0.224
- Valid Host Range: 172.16.0.225 – 172.16.0.238
- Broadcast Address: 172.16.0.239
- Usable Hosts: 14

Final Allocation Table:

Departme nt	Prefi x	Network Address	Valid Host Range	Broadca st	Usable Hosts
HR	/25	172.16.0.0	.1 – .126	.127	126
Sales	/26	172.16.0.128	.129 – .190	.191	62
IT	/27	172.16.0.192	.193 – .222	.223	30
Admin	/28	172.16.0.224	.225 – .238	.239	14

Remaining Unused IP Address Range:

The last allocated address is 172.16.0.239.

Therefore, the remaining unused range is:

172.16.0.240 – 172.16.255.255

This includes:

$65536 - (128 + 64 + 32 + 16) = 65296$
 $65536 - (128 + 64 + 32 + 16) = 65296$

Total unused addresses = 65,296

Conclusion: The departments are efficiently allocated non-overlapping subnets, while a large portion of the /16 network remains available for future expansion.

4. Subnetting 172.16.10.0/24 Using /27

Given:

- Original Network: 172.16.10.0/24
- New Subnet Prefix: /27

Step 1: Number of Subnets

The original prefix is /24 and the new prefix is /27.

Bits borrowed:

$$27 - 24 = 3$$

Number of subnets:

$$2^3 = 8$$

Therefore, **8 subnets** are created.

Step 2: Usable Hosts Per Subnet

Host bits:

$$32 - 27 = 5$$

Total addresses:

$$2^5 = 32$$

Usable hosts:

$$32 - 2 = 30$$

Therefore, each subnet has **30 usable host addresses**.

Step 3: List the Subnets

The /27 block size is:

$$256-224=32 \quad 256-224=32 \quad 256-224=32$$

The subnets are:

1. 172.16.10.0 – 31
2. 172.16.10.32 – 63
3. 172.16.10.64 – 95
4. 172.16.10.96 – 127
5. 172.16.10.128 – 159
6. 172.16.10.160 – 191
7. 172.16.10.192 – 223
8. 172.16.10.224 – 255

Step 4: Find the Subnet of 172.16.10.78

The value 78 lies between 64 and 95.

Therefore:

- Network Address: 172.16.10.64
- Valid Host Range: 172.16.10.65 – 172.16.10.94
- Broadcast Address: 172.16.10.95

The IP 172.16.10.78 is a valid host IP in this subnet.

Step 5: Check 172.16.10.95

172.16.10.95 is the **broadcast address** of the subnet.

Therefore, it belongs to the same subnet range but **cannot be assigned to a host**.

Final Answer:

- Number of Subnets = 8
- Usable Hosts per Subnet = 30
- Subnet containing 172.16.10.78 = 172.16.10.64/27
- Network Address = 172.16.10.64
- Valid Host Range = 172.16.10.65 – 172.16.10.94
- Broadcast Address = 172.16.10.95
- 172.16.10.78 = Valid host address
- 172.16.10.95 = Same subnet, but broadcast address

Conclusion: The IP 172.16.10.78 is a valid host in the 172.16.10.64/27 subnet, whereas .95 is reserved as the broadcast address.

5. IPv6 Address Compression and /64 Network Prefix

Given IPv6 Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 1: Remove Leading Zeros

In IPv6 notation, leading zeros in each hexadecimal group can be removed.

Original:

2001:0db8:0000:0000:0000:ff00:0042:8329

After removing leading zeros:

2001:db8:0:0:0:ff00:42:8329

Step 2: Compress Consecutive Zero Groups

The three consecutive 0 groups can be replaced by ::.

Therefore, the shortest compressed form is:

2001:db8::ff00:42:8329

This is the correct compressed IPv6 notation.

Step 3: Expand the Compressed Address

Compressed address:

2001:db8::ff00:42:8329

The :: represents three zero groups.

Therefore, the expanded form is:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 4: Determine the /64 Network Prefix

An IPv6 /64 prefix uses the first 64 bits, which means the first four hexadecimal groups.

Given:

2001:0db8:0000:0000:ff00:0042:8329

First four groups:

2001:0db8:0000:0000

Therefore, the network prefix is:

2001:0db8:0000:0000::/64

In compressed form:

2001:db8::/64

The host/interface identifier is:

0000:ff00:0042:8329

Step 5: Calculate Total Possible Host Addresses

For a /64 network:

$128 - 64 = 64$

Therefore, there are **64 host bits**.

Total possible addresses:

2^{64}
 $= 18,446,744,073,709,551,616$

Thus, the network contains:

18,446,744,073,709,551,616 possible IPv6 addresses.

Final Answer:

Parameter

Result

Original IPv6	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed IPv6	2001:db8::ff00:42:8329
Expanded IPv6	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix Length	/64
Network Prefix	2001:db8::/64
Host Bits	64
Total Possible Addresses	$2^{64} = 18,446,744,073,709,551,616$

Conclusion: IPv6 compression reduces the address length for easier representation, while the /64 prefix divides the address into a **64-bit network prefix** and a **64-bit interface identifier**. The /64 network provides an extremely large address space suitable for modern IPv6 network design.

